

Import Standard Libraries

```
In [7]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

import warnings
warnings.filterwarnings('ignore')
```

Read data

```
In [8]: df_Swiggy = pd.read_csv('Swiggy Bangalore Outlet Details.csv')
df_Swiggy.head()
```

```
Out[8]:
```

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	₹ 150
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	₹ 400
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	₹ 126
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	₹ 400
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	₹ 450

```
In [9]: # how many features are there in datasets
df_Swiggy.columns
```

```
Out[9]: Index(['Shop_Name', 'Cuisine', 'Location', 'Rating', 'Cost_for_Two'], dtype='object')
```

```
In [10]: # check for missing values in dataset
df_Swiggy.isnull().sum()
```

```
Out[10]: Shop_Name      0
Cuisine      0
Location     0
Rating       0
Cost_for_Two 0
dtype: int64
```

```
In [11]: df_Swiggy.info() #80%---cleaning

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 118 entries, 0 to 117
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   Shop_Name   118 non-null   object
 1   Cuisine     118 non-null   object
 2   Location    118 non-null   object
 3   Rating      118 non-null   object
 4   Cost_for_Two 118 non-null   object
dtypes: object(5)
memory usage: 4.7+ KB
```

```
In [12]: df_Swiggy.describe(include="all")
```

```
Out[12]:
```

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
count	118	118	118	118	118
unique	115	79	65	13	30
top	La Pino'z Pizza	North Indian	BTM, BTM	4.1	₹ 300
freq	2	12	13	30	16

Data Pre-Processing & Cleaning

```
In [13]: df_Swiggy.duplicated().sum()
```

```
Out[13]: 0
```

```
In [14]: df_Swiggy.head()
```

Out[14]:	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	₹ 150
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	₹ 400
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	₹ 126
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	₹ 400
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	₹ 450

```
In [15]: # how many different ratings are in dataset
df_Swiggy['Rating'].unique()
```

```
Out[15]: array(['4.3', '4.4', '4.1', '4.2', '3.9', '3.8', '4', '3.7', '3.6', '4.8',
        '4.5', '4.6', '--'], dtype=object)
```

```
In [16]: # a=np.array([1,2.3,'3']) #upcasting
```

```
In [2]: s1="₹ 450"
s1.split()[1]
```

```
Out[2]: '450'
```

```
In [3]: s1="₹ 450"
s1.strip("₹ ")
```

```
Out[3]: '450'
```

```
In [ ]:
```

```
In [19]: # Replace '--' rating with zero(0)
df_Swiggy['Rating'] = df_Swiggy['Rating'].str.replace('--', '0').astype(float)
```

```
In [20]: # How many unique entries for 'Cost_for_Two' feature
df_Swiggy['Cost_for_Two'].unique()
```

```
Out[20]: array(['₹ 150', '₹ 400', '₹ 126', '₹ 450', '₹ 350', '₹ 200', '₹ 500',
        '₹ 247', '₹ 550', '₹ 300', '₹ 129', '₹ 250', '₹ 268', '₹ 600',
        '₹ 527', '₹ 130', '₹ 257', '₹ 280', '₹ 399', '₹ 220', '₹ 800',
        '₹ 100', '₹ 178', '₹ 120', '₹ 251', '₹ 650', '₹ 132', '₹ 153',
        '₹ 193'], dtype=object)
```

```
In [21]: df_Swiggy['Cost_for_Two']=df_Swiggy['Cost_for_Two'].apply(lambda x:int(x.strip("₹ ")))
```

```
In [22]: df_Swiggy['Cost_for_Two'].dtype
```

```
Out[22]: dtype('int64')
```

```
In [23]: # # Remove the rupees symbol (₹) from the Cost_for_Two  
# df_Swiggy['Cost_for_Two'] = df_Swiggy['Cost_for_Two'].apply(lambda string: string.split(' ')[1]).astype(int)
```

```
In [24]: df_Swiggy.head()
```

```
Out[24]:
```

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	150
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	400
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	126
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	400
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	450

```
In [25]: df_Swiggy['Cost_for_Two'].unique()
```

```
Out[25]: array([150, 400, 126, 450, 350, 200, 500, 247, 550, 300, 129, 250, 268,  
        600, 527, 130, 257, 280, 399, 220, 800, 100, 178, 120, 251, 650,  
        132, 153, 219, 193], dtype=int64)
```

```
In [26]: df_Swiggy.head()
```

```
Out[26]:
```

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	150
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	400
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	126
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	400
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	450

```
In [27]: df_Swiggy.info()
```

```
Out[27]: Shop_Name      object
Cuisine      object
Location      object
Rating      float64
Cost_for_Two  int64
dtype: object
```

```
In [29]: df_Swiggy.describe()
```

```
Out[29]:
```

	Rating	Cost_for_Two
count	118.000000	118.000000
mean	4.061864	321.008475
std	0.430845	137.286804
min	0.000000	100.000000
25%	4.000000	204.750000
50%	4.100000	300.000000
75%	4.300000	400.000000
max	4.800000	800.000000

```
In [30]: df_Swiggy.head()
```

```
Out[30]:
```

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	150
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	400
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	126
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	400
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	450

```
In [31]: ##Distribution of 'Ratings':
```

```
Loading [MathJax]/extensions/Safe.js s = df_Swiggy[df_Swiggy['Rating'] > 0]
```

```
df_valid_Ratings
```

```
Out[31]:
```

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	150
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	400
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	126
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	400
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	450
...	...	...	...	...	...
113	Wok Paper Scissors	Pan-Asian, Chinese, Asian	JNC Road, Koramangala	3.9	219
114	Savoury Restaurant	Arabian, Middle Eastern, North Indian, Grill, ...	Madiwala, BTM	4.1	600
115	Royal Treat	North Indian, Chinese, Seafood, Biryani	5th block Koramangala, Koramangala	4.2	193
116	Thali 99	North Indian	Koramangala, Koramangala	4.3	200
117	Mani's Dum Biryani	Andhra, Biryani	1st Block, Koramangala	4.2	400

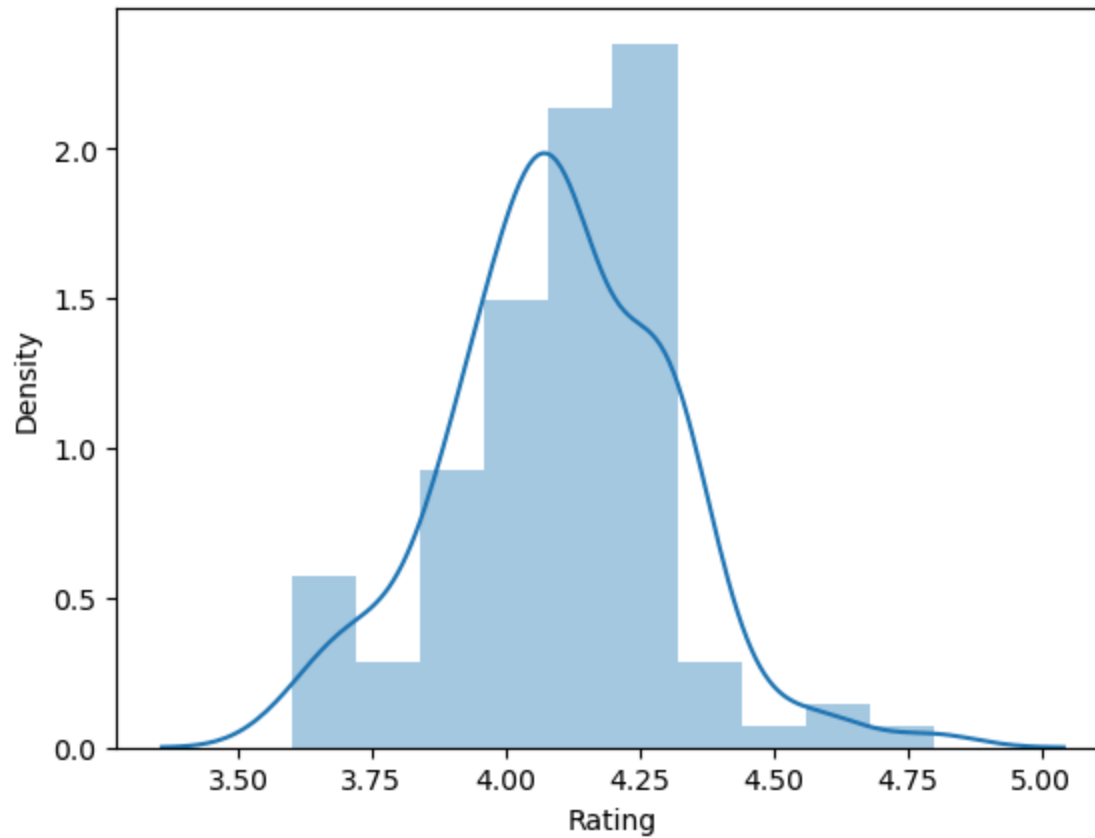
117 rows × 5 columns

```
In [32]:
```

```
#Distribution of 'Rating':  
sns.distplot(df_valid_Ratings['Rating'])
```

```
Out[32]:
```

```
<Axes: xlabel='Rating', ylabel='Density'>
```



Conclusion:

- From this **Distribution Plot**, We can conclude that **More than '50%' of Restaurants** are having a **Rating** greater than "**4.1**" with a **Maximum Rating** of "**4.8**" which is considered as a decent Rating.
- And It also means that, Most of these Restaurants are doing very well & Rated accordingly by the Customers.

```
In [33]: # Handling Feature --> Location
df_Swiggy['Location'].unique()
```

```
Out[33]: array(['Koramangala, Koramangala', 'Sector 5, HSR',
        '6th Block, Koramangala', 'HSR, HSR', '5th Block, Koramangala',
        'Koramangala 4th Block, Koramangala', 'BTM 2nd Stage, BTM',
        'BTM, BTM', '9th Main road, Koramangala', 'outer ring road, BTM',
        '7th Block, Koramangala', '1st Main, Koramangala',
        'Bommanahalli, BTM', '6th block, Koramangala', 'Sector 4, HSR',
        'BTM 1st stage, BTM', 'Jakkasandra Extn, Koramangala',
        'Marutnagar Main Road, BTM', '1st Block, Koramangala',
        '4th Cross, BTM', 'koramangala, Koramangala', 'BTM 2nd stage, BTM',
        '3rd main, BTM', 'HSR 1st sector, HSR', 'Sector 7, HSR',
        '3rd Sector, HSR', 'Chocolate Factory Road, BTM',
        '16th Main Road, 2nd Stage, BTM', '1st Stage, BTM',
        'Hosur Main Road, Koramangala',
        '1st Cross Road, 5th Block, Near Jyothi Nivas College, Koramangala',
        'Mico Layout, BTM', '4th Cross, Koramangala',
        '4th Block, Koramangala', 'Intermediate Ring Road, Koramangala',
        '3rd sector, HSR', '8TH BLOCK, Koramangala',
        '4th b cross, Koramangala', 'SG palaya, BTM',
        'Venkatapura Main Rd, Teacher's Colony, Jakkasandra, HSR',
        'KHB Colony, Koramangala', 'Sector 3, HSR',
        'Bannerghatta Road, Jayanagar',
        '80 Feet Peripheral Road, Koramangala', 'Btm, BTM',
        'Near Wipro Park Signal, Koramangala', '16th Main Road, BTM',
        '2nd Stage, BTM', 'Kuvempu Nagar, Stage 2, BTM',
        'Koramangala 1st block, Koramangala',
        '5th Block Kormangala, Koramangala', 'Koramangla, Koramangala',
        '5th block, Koramangala', '9th Main Rd, Sector 6, HSR Layout, HSR',
        'Jay Bheema Nagar, BTM', 'Koramangala 6th block, Koramangala',
        'Maruthi Nagar, BTM', 'Sector 6, HSR',
        'Jakkasandra Village, Koramangala', '4th block, Koramangala',
        'Madiwala Junction, BTM', 'kormangala, Koramangala',
        'JNC Road, Koramangala', 'Madiwala, BTM',
        '5th block Koramangala, Koramangala'], dtype=object)
```

```
In [34]: df_Swiggy['Location'].nunique() # number of unique Locations
```

```
Out[34]: 65
```

```
In [35]: # Locations that contains 'Koramangala'
         swiggy_Koramangala = df_Swiggy[df_Swiggy['Location'].str.contains('Koramangala')]
         swiggy_Koramangala
```

Out[35]:

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	150
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	126
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	450
5	Kitchens of Punjab	North Indian	Koramangala 4th Block, Koramangala	4.2	350
9	Yumlane Pizza	Pizzas, Italian, Mexican	9th Main road, Koramangala	3.8	150
...	...	...	...	...	...
112	Kritunga	Andhra, Biryani	5th Block, Koramangala	3.9	500
113	Wok Paper Scissors	Pan-Asian, Chinese, Asian	JNC Road, Koramangala	3.9	219
115	Royal Treat	North Indian, Chinese, Seafood, Biryani	5th block Koramangala, Koramangala	4.2	193
116	Thali 99	North Indian	Koramangala, Koramangala	4.3	200
117	Mani's Dum Biryani	Andhra, Biryani	1st Block, Koramangala	4.2	400

64 rows × 5 columns

```
In [36]: # Locations that contains 'HSR'
swiggy_HSR = df_Swiggy[df_Swiggy['Location'].str.contains('HSR')]
swiggy_HSR
```

Out[36]:

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	400
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	400
8	Hotel Manu	South Indian, Kerala, Chinese, North Indian	HSR, HSR	4.1	350
19	Shree Khana Khazana	Indian, Rajasthani	Sector 4, HSR	4.1	350
24	New Udupi Grand	Chinese, Jain, North Indian, South Indian	HSR, HSR	4.3	150
36	Biriyani Zone	North Indian, Chinese, Biryani	HSR 1st sector, HSR	4.1	600
37	Gongura's	North Indian, Chinese, Biryani	Sector 7, HSR	3.8	300
39	Leon Grill	Turkish, Portuguese, American	3rd Sector, HSR	4.3	300
41	Cakewala	Desserts	HSR, HSR	4.3	450
57	Donne Biriyani House	South Indian	3rd sector, HSR	4.0	300
58	Nanda's	Andhra, Biryani	HSR, HSR	4.0	400
61	Cake Garden	Desserts, Bakery	HSR, HSR	3.9	250
71	Nizams Biryani	Biryani, Juices, Kebabs	Venkatapura Main Rd, Teacher's Colony, Jakkasa...	3.6	200
73	Punjabi Rasoi	North Indian	Sector 3, HSR	4.0	800
98	Mandya Gowdru Donne Biryani	Biryani	HSR, HSR	0.0	350
99	Dindigul Thalapakatti Biryani	North Indian	HSR, HSR	4.1	650
101	Easy Bites	Snacks, American	9th Main Rd, Sector 6, HSR Layout, HSR	3.8	200
107	Junior Kuppanna	Chettinad, South Indian	Sector 6, HSR	4.0	550

In [37]:

```
# Locations that contains 'BTM'
swiggy_BTMT = df_Swiggy[df_Swiggy['Location'].str.contains('BTM')]
swiggy_BTMT
```

Out[37]:

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
6	99 VARIETY DOSA AND PAV BHAJI- Malli Mane Food...	Fast Food, North Indian, Chinese	BTM 2nd Stage, BTM	4.1	200
7	La Pino'z Pizza	Italian	BTM, BTM	3.9	500
10	Ambur Star Briyani	Chinese, South Indian, North Indian, Desserts,...	outer ring road, BTM	4.1	500
17	Sri Lakshmi Dhaba	North Indian	Bommanahalli, BTM	3.7	200
20	Just Bake - Cakes & confectioners	Desserts, Bakery	BTM 1st stage, BTM	4.3	300
22	Hotel Godavari	North Indian, Chinese, Hyderabadi	Marutinagar Main Road, BTM	4.0	400
25	Swad Punjab da	Indian	BTM, BTM	4.1	250
27	High N Hungry	Andhra, Biryani, Chinese, Desserts, Fast Food,...	4th Cross, BTM	4.1	350
31	Bengali Fun Foods	North Indian	BTM 2nd stage, BTM	4.2	300
33	Oottupura	Kerala, South Indian	BTM, BTM	4.3	268
35	Hyderabadi Biryani Hub	North Indian, Chinese, Biryani	3rd main, BTM	3.9	450
40	Venu's Donne Biryani	Biryani	Chocolate Factory Road, BTM	4.3	300
42	Swadista Aahar	South Indian, Snacks, North Indian, Chinese	16th Main Road, 2nd Stage, BTM	4.1	250
44	Svadu Pure Ghee Sweets	Desserts, Fast Food, Sweets, Chaat	1st Stage, BTM	4.1	200
45	Sai Abhiruchi	Chinese, South Indian, Andhra, Hyderabadi	BTM, BTM	3.7	250
49	Balaji's Veg	North Indian, Chinese, South Indian	Mico Layout, BTM	4.1	300
51	Donne Biryani Mandi	Biryani, Andhra, South Indian	BTM, BTM	4.0	150
60	calicut cafe restaurant	Fast Food, Beverages	BTM, BTM	4.1	280
65	World of asia	Beverages, Chinese	BTM, BTM	4.0	250
66	Ghar Ka Khana	North Indian	BTM, BTM	4.2	220
68	KANNUR FOOD POINT	Kerala, Chinese	SG palaya, BTM	3.9	300
69	KANNOOR RESTAURANT	North Indian, Chinese	BTM, BTM	4.0	250

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
70	Fattoush	Arabian, Beverages, Biryani, Chinese, Desserts...	BTM, BTM	3.9	400
76	BIRIYANI TASTE MASTH(BTM)	North Indian, South Indian	Btm, BTM	4.2	300
79	Tandoori Merchant	Andhra, Biryani, Chinese, Desserts, Fast Food,...	4th Cross, BTM	4.2	100
80	Chinese Bae	Chinese, Thai	BTM, BTM	4.5	450
83	Abhiruchi Hotel	Chinese, Hyderabad, Biryani, Indian, South In...	BTM, BTM	4.0	250
84	Punjabi Swag	Punjabi, North Indian, Chinese, Fast Food, Hea...	16th Main Road, BTM	3.7	400
86	Gyaani Da Punjabi Dhaba	North Indian	2nd Stage, BTM	4.0	500
87	Biriyani Bhatti	Biryani, Hyderabad, Andhra, North Indian, Sou...	Kuvempu Nagar, Stage 2, BTM	4.1	350
92	BIRYANI CRAFTS	Indian	BTM, BTM	4.1	500
104	R.B Food Point	Chinese, North Indian	Jay Bheema Nagar, BTM	3.7	350
106	New Tasty Cafeteria	Andhra, Chettinad, Chinese, Mughlai, North Indian	Maruthi Nagar, BTM	4.0	350
110	Biryani Pot	North Indian, Biryani	Madiwala Junction, BTM	4.0	500
114	Savoury Restaurant	Arabian, Middle Eastern, North Indian, Grill, ...	Madiwala, BTM	4.1	600

As we can see the Returants are given from only 3 Locations.

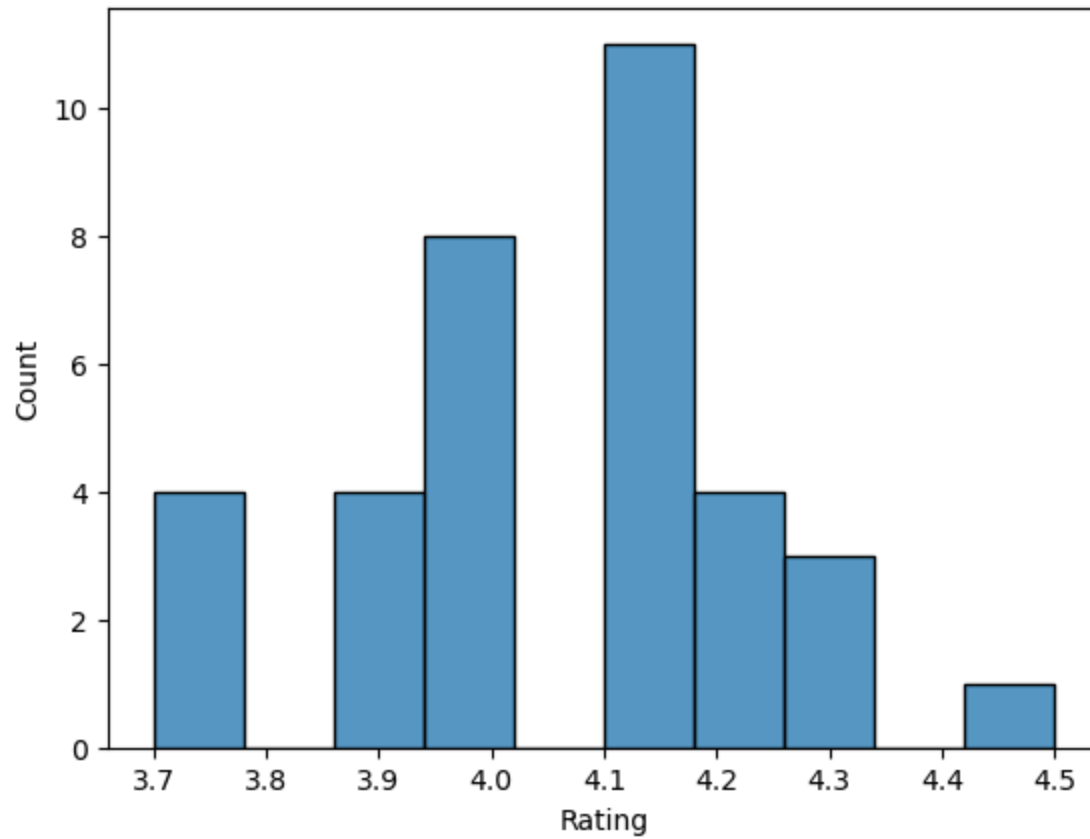
Area-wise Analysis -

BTM Area

```
In [38]: sns.histplot(swiggy_BTMTM['Rating'], bins = 10)
```

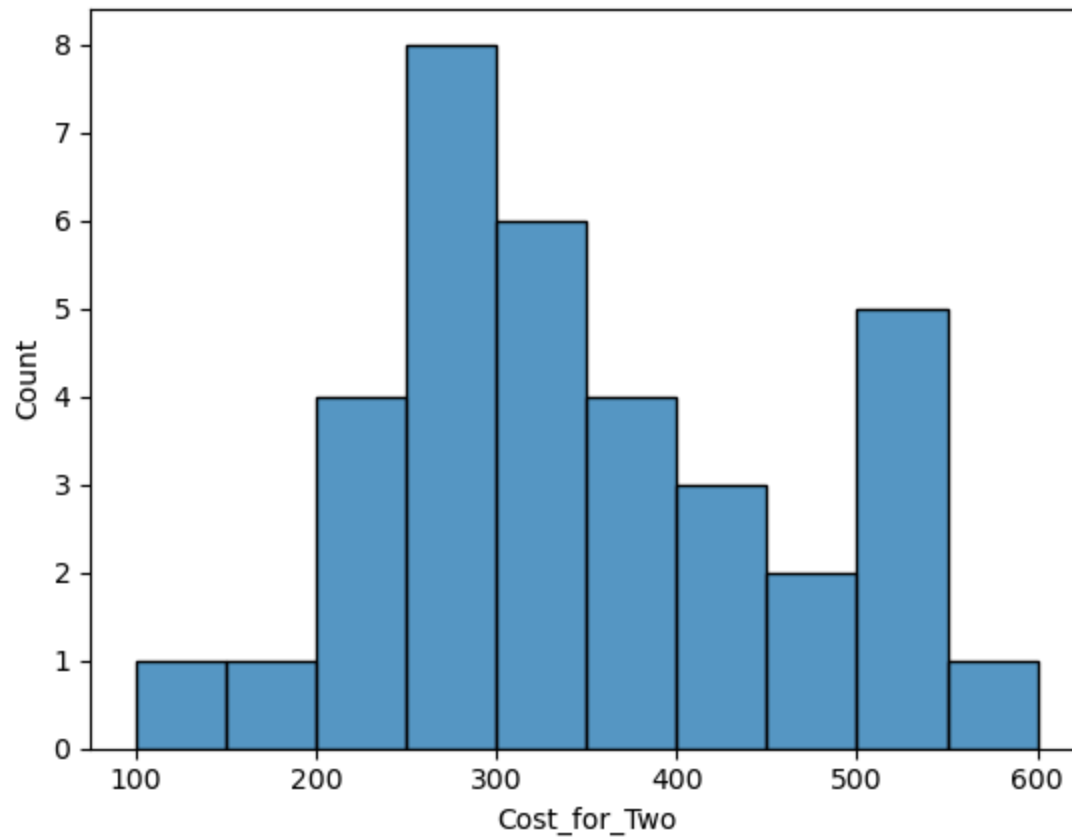
```
plt.xlabel('Rating', ylabel='Count')>
```

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```
In [39]: sns.histplot(swiggy_BTM['Cost_for_Two'], bins = 10)
```

```
Out[39]: <Axes: xlabel='Cost_for_Two', ylabel='Count'>
```



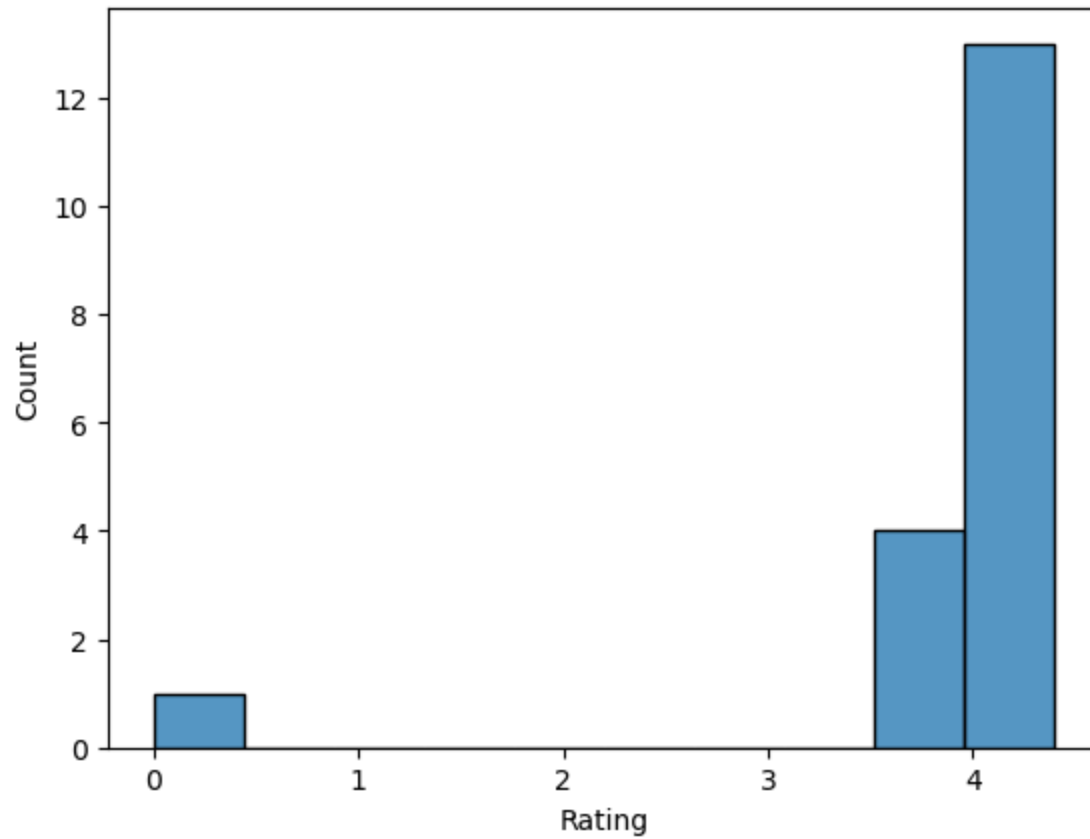
Conclusion:

- BTM : Most has 4.0 to 4.2 Rating and Approx. Cost for Two People lies between 200 to 350. (Max. Cost goes upto 600)

HSR Area

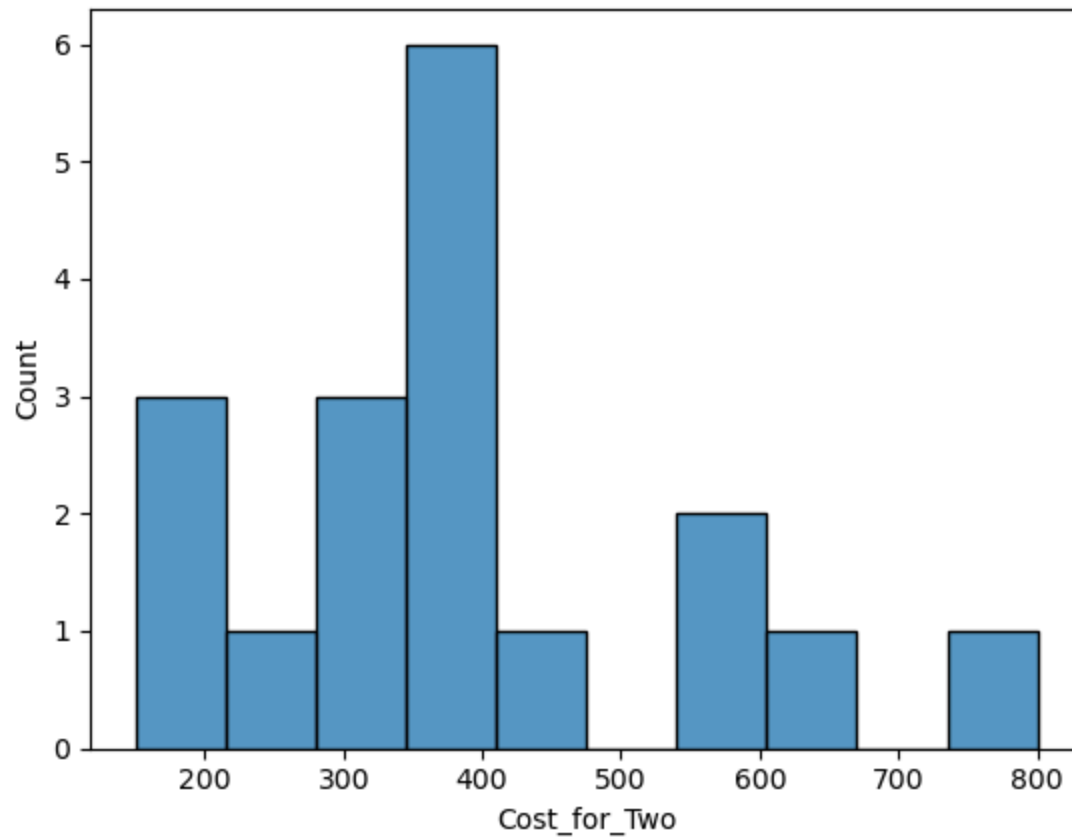
```
In [40]: sns.histplot(swiggy_HSR['Rating'], bins = 10)
```

```
Out[40]: <Axes: xlabel='Rating', ylabel='Count'>
```



```
In [41]: sns.histplot(swiggy_HSR['Cost_for_Two'], bins = 10)
```

```
Out[41]: <Axes: xlabel='Cost_for_Two', ylabel='Count'>
```



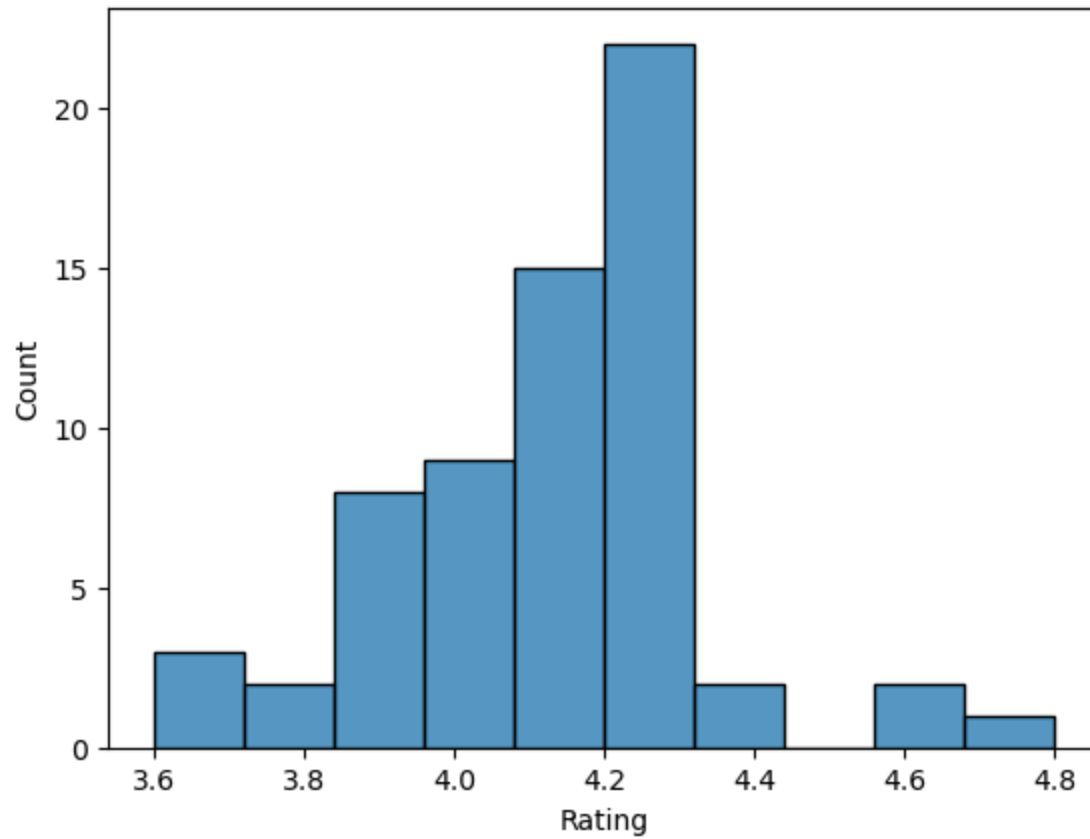
Conclusion:

- HSR : Most has 4 or above Rating and Approx. Cost for Two People lies between 300 to 400. (Max. Cost goes upto 800)

Koramangala Area

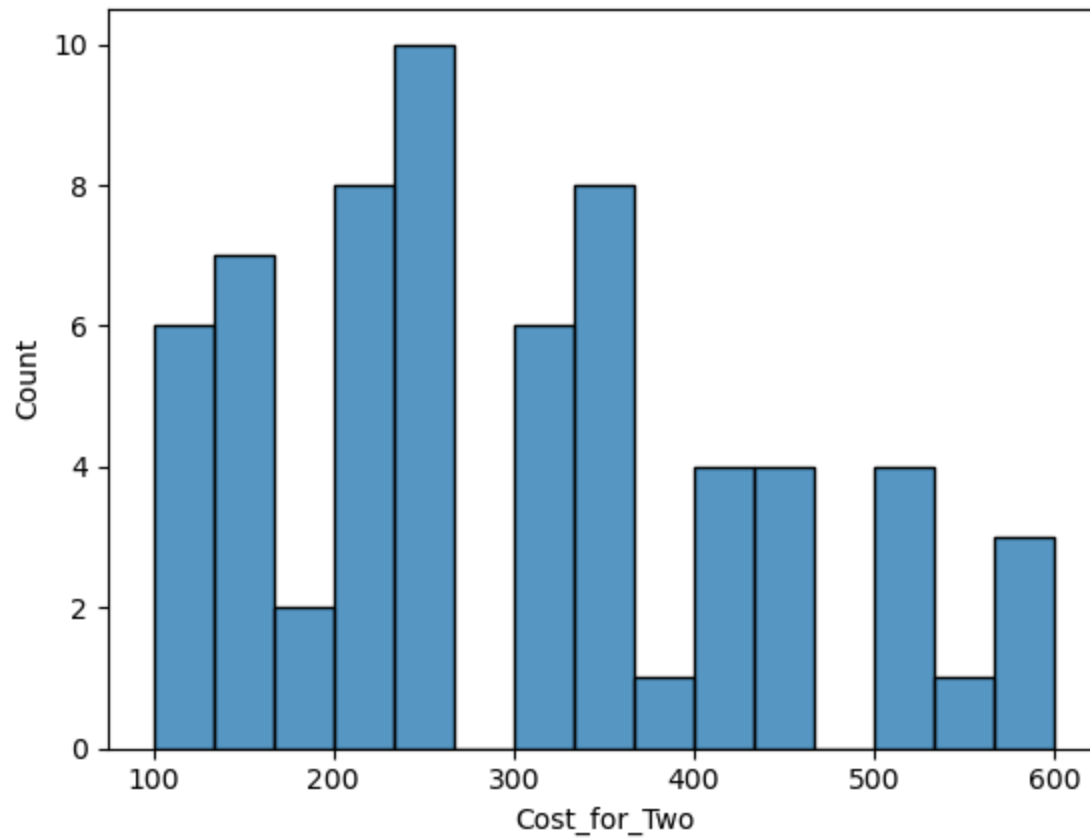
```
In [42]: sns.histplot(swiggy_Koramangala["Rating"], bins=10)
```

```
Out[42]: <Axes: xlabel='Rating', ylabel='Count'>
```



```
In [43]: sns.histplot(swiggy_koramangala["Cost_for_Two"], bins=15)
```

```
Out[43]: <Axes: xlabel='Cost_for_Two', ylabel='Count'>
```



Conclusion:

- Koramangala : Most has 4.0 to 4.3 Rating and Approx. Cost for Two People lies between 200 to 350. (Max. Cost goes upto 600)

Conclusion of Areawise Analysis

With this we can conclude the Most Costly Area is HSR.

```
In [44]: df_Swiggy.head()
```

Out[44]:

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	150
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	400
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	126
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	400
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	450

In [45]: *# Q. Analyse "Approx Cost of 2 People" vs "Rating". Find out the relationship between them.*
df_Highest_Rated_Restaurants = df_Swiggy[df_Swiggy["Rating"] >= 4.0]
df_Highest_Rated_Restaurants

Out[45]:

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	150
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	400
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	126
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	400
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	450
...	...	...	...	...	...
111	Bowl 99	North Indian, South Indian	kormangala, Koramangala	4.4	200
114	Savoury Restaurant	Arabian, Middle Eastern, North Indian, Grill, ...	Madiwala, BTM	4.1	600
115	Royal Treat	North Indian, Chinese, Seafood, Biryani	5th block Koramangala, Koramangala	4.2	193
116	Thali 99	North Indian	Koramangala, Koramangala	4.3	200
117	Mani's Dum Biryani	Andhra, Biryani	1st Block, Koramangala	4.2	400

92 rows × 5 columns

In [46]: df_Highest_Rated_Restaurants = df_Highest_Rated_Restaurants.loc[:, ["Shop_Name", "Rating", "Cost_for_Two"]]
df_Highest_Rated_Restaurants

Out[46]:

	Shop_Name	Rating	Cost_for_Two
0	Kanti Sweets	4.3	150
1	Mumbai Tiffin	4.4	400
2	Sri Krishna sagar	4.1	126
3	Al Daaz	4.4	400
4	Beijing Bites	4.1	450
...	...	...	...
111	Bowl 99	4.4	200
114	Savoury Restaurant	4.1	600
115	Royal Treat	4.2	193
116	Thali 99	4.3	200
117	Mani's Dum Biryani	4.2	400

92 rows × 3 columns

In [47]:

```
# Grouping on Shop_Name and Rating
df_Highest_Rated_Restaurants = df_Highest_Rated_Restaurants.
    groupby(["Shop_Name", "Rating"])["Cost_for_Two"].agg("mean")
df_Highest_Rated_Restaurants = df_Highest_Rated_Restaurants.reset_index()
df_Highest_Rated_Restaurants
```

Cell In[47], line 2

```
df_Highest_Rated_Restaurants = df_Highest_Rated_Restaurants.
```

SyntaxError: invalid syntax

In [48]:

```
#df_Highest_Rated_Restaurants.groupby(["Shop_Name", "Rating"])["Cost_for_Two"].agg("mean").reset_index()
```

In [49]:

```
import plotly.express as px
fig = px.scatter(
    x=df_Highest_Rated_Restaurants["Cost_for_Two"],
    y=df_Highest_Rated_Restaurants["Rating"],
    color=df_Highest_Rated_Restaurants["Rating"],
    df_Highest_Rated_Restaurants["Cost_for_Two"],
    labels={
```

```

        "x": "Approx. Cost_for_Two",
        "y": "Rating",
        "color": "Rating_Indicator"
    }
)
fig.update_layout(
    template="plotly_dark",
    title="Analyse 'Approx Cost of 2 People' vs 'Rating'"
)
fig.show()

```



Conclusion:

- From this '**Scatter Plot**', We can clearly say that - Bigger the Bubble, Higher the Price is... Similarly, Smaller the Bubble, Lesser the Price is...
- Furthermore, We can also say that, **Most of the Affordable/Budgeted Restaurants** are having **Excellent Rating** as well.
 - Same we can see, For **Approx. Cost** of "200", "150", "250", and "450", the **Ratings** were "4.8", "4.6", and "4.5" respectively.
 - This might be because **Most of the people prefer Affordable/Budget-Restaurants** which also provides **good quality of Cuisines**.
- and On the other hand, There are **few Expensive Restaurants** who **doesn't** have that much **Rating** and they are **Expensive**.
 - **Restaurants** which **Costs** around "600" to "800" for **Two People** are having the **Ratings in between '4.0' to '4.1'** which is **too less** as compared to **Affordable/Budgeted Restaurants**.
- So, That's a Conclusion we can draw from this '**Scatter Plot**'.

```
In [39]: ##Q. Analyze Affordable/Budgeted and Highest Rated Restaurants of Bangalore:
df_Affordable_Restaurants = df_Swiggy[
    (df_Swiggy["Cost_for_Two"] <= 500) & (df_Swiggy["Rating"] >= 4.0)
]
df_Affordable_Restaurants
```

Out[39]:

	Shop_Name	Cuisine	Location	Rating	Cost_for_Two (₹)
0	Kanti Sweets	Sweets	Koramangala, Koramangala	4.3	150
1	Mumbai Tiffin	North Indian, Home Food, Thalís, Combo	Sector 5, HSR	4.4	400
2	Sri Krishna sagar	South Indian, North Indian, Fast Food, Beverag...	6th Block, Koramangala	4.1	126
3	Al Daaz	American, Arabian, Chinese, Desserts, Fast Foo...	HSR, HSR	4.4	400
4	Beijing Bites	Chinese, Thai	5th Block, Koramangala	4.1	450
...	...	...	...	...	...
110	Biryani Pot	North Indian, Biryani	Madiwala Junction, BTM	4.0	500
111	Bowl 99	North Indian, South Indian	kormangala, Koramangala	4.4	200
115	Royal Treat	North Indian, Chinese, Seafood, Biryani	5th block Koramangala, Koramangala	4.2	193
116	Thali 99	North Indian	Koramangala, Koramangala	4.3	200
117	Mani's Dum Biryani	Andhra, Biryani	1st Block, Koramangala	4.2	400

82 rows × 5 columns

In [40]:

```
df_Affordable_Restaurants = df_Affordable_Restaurants.groupby(["Shop_Name", "Rating"])["Cost_for_Two"].agg("mean")
df_Affordable_Restaurants = df_Affordable_Restaurants.reset_index()
df_Affordable_Restaurants
```

Out[40]:

	Shop_Name	Rating	Cost_for_Two (₹)
0	99 VARIETY DOSA AND JUICE-Malli mane food court	4.1	100.0
1	99 VARIETY DOSA AND PAV BHAJI- Malli Mane Food...	4.1	200.0
2	A2B - Adyar Ananda Bhavan	4.2	450.0
3	Abhiruchi Hotel	4.0	250.0
4	Al Daaz	4.4	400.0
...	...	...	...
76	Venu's Donne Biryani	4.3	300.0
77	WarmOven Cake & Desserts	4.1	200.0
78	World of asia	4.0	250.0
79	XO Belgian Waffle	4.3	250.0
80	calicut cafe restaurant	4.1	280.0

81 rows × 3 columns

```
In [41]: df_Affordable_Restaurants.sort_values(by=["Rating"], ascending=False, inplace=True)
df_Affordable_Restaurants
```

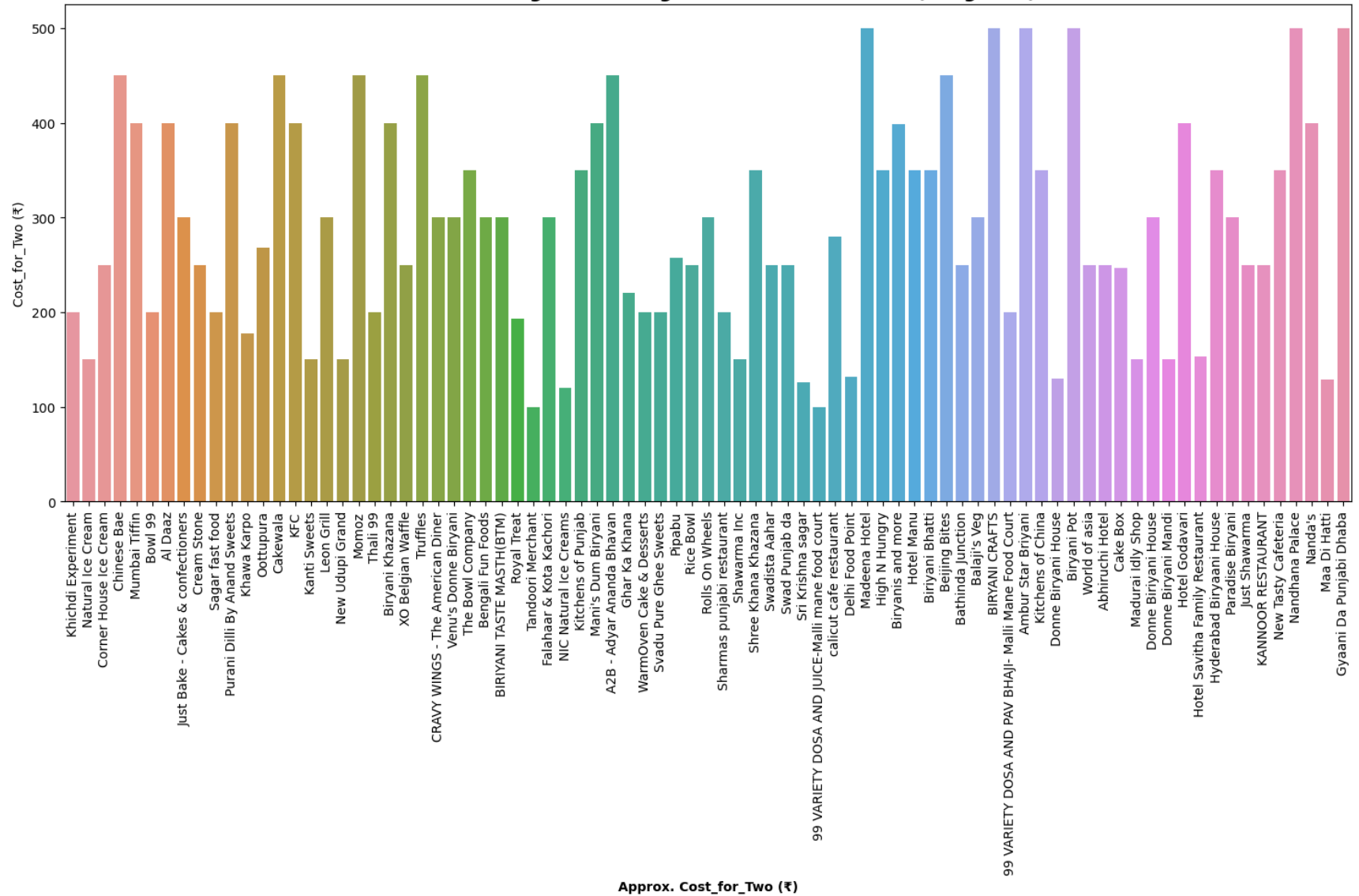
Out[41]:

	Shop_Name	Rating	Cost_for_Two (₹)
41	Khichdi Experiment	4.8	200.0
54	Natural Ice Cream	4.6	150.0
21	Corner House Ice Cream	4.6	250.0
20	Chinese Bae	4.5	450.0
50	Mumbai Tiffin	4.4	400.0
...	...	...	...
55	New Tasty Cafeteria	4.0	350.0
53	Nandhana Palace	4.0	500.0
52	Nanda's	4.0	400.0
45	Maa Di Hatti	4.0	129.0
29	Gyaani Da Punjabi Dhaba	4.0	500.0

81 rows × 3 columns

```
In [42]: plt.figure(figsize=(18, 7))
sns.barplot(
    x=df_Affordable_Restaurants["Shop_Name"],
    y=df_Affordable_Restaurants["Cost_for_Two"],
    data=df_Affordable_Restaurants
)
plt.title(
    "Affordable/Budgeted and Highest Rated Restaurants (Bangalore)",
    fontsize=14,
    fontweight="bold",
    fontstyle="italic"
)
plt.xlabel("Shop_Name", fontsize=10, fontweight="bold")
plt.ylabel("Approx. Cost_for_Two", fontsize=10, fontweight="bold")
plt.xticks(rotation=90)
plt.show()
```

Affordable/Budgeted and Highest Rated Restaurants (Bangalore)



```
In [43]: # Q. Top 15 Cheapest & Highest Rated Restaurants with Approx. Cost for 2 People:
df_Cheapest_Restaurants = df_Affordable_Restaurants.sort_values(
    by="Cost_for_Two", ascending=True
)
df_Cheapest_Restaurants
```

Out[43]:

	Shop_Name	Rating	Cost_for_Two (₹)
72	Tandoori Merchant	4.2	100.0
0	99 VARIETY DOSA AND JUICE-Malli mane food court	4.1	100.0
51	NIC Natural Ice Creams	4.2	120.0
68	Sri Krishna sagar	4.1	126.0
45	Maa Di Hatti	4.0	129.0
...	...	...	...
5	Ambur Star Briyani	4.1	500.0
7	BIRYANI CRAFTS	4.1	500.0
53	Nandhana Palace	4.0	500.0
46	Madeena Hotel	4.1	500.0
29	Gyaani Da Punjabi Dhaba	4.0	500.0

81 rows × 3 columns

```
In [44]: fig = px.bar(
    data_frame=df_Cheapest_Restaurants,
    x=df_Cheapest_Restaurants["Shop_Name"][0:15],
    y=df_Cheapest_Restaurants["Cost_for_Two"][0:15],
    color=df_Cheapest_Restaurants["Rating"][0:15],
    labels={
        "x": "Restaurant_Name",
        "y": "Approx. Cost_for_Two (₹)",
        "color": "Rating",
    }
)
fig.update_layout(
    template="plotly_dark",
    title="Top 15 Cheapest & Highest Rated Restaurants with Approx. Cost for 2 People",
)

fig.show()
```



<Figure size 640x480 with 0 Axes>

```
In [45]: # Q. Top 15 Expensive & Highest Rated Restaurants with Approx. Cost for 2 People:
df_Expensive_Restaurants = df_Highest_Rated_Restaurants.sort_values(
    by="Cost_for_Two", ascending=False)
df_Expensive_Restaurants
```

Out[45]:

	Shop_Name	Rating	Cost_for_Two (₹)
67	Punjabi Rasoi	4.0	800.0
26	Dindigul Thalapakatti Biryani	4.1	650.0
73	Savoury Restaurant	4.1	600.0
81	Taco Bell	4.3	600.0
66	Pizza Hut	4.0	600.0
...	...	...	...
49	Maa Di Hatti	4.0	129.0
77	Sri Krishna sagar	4.1	126.0
56	NIC Natural Ice Creams	4.2	120.0
82	Tandoori Merchant	4.2	100.0
0	99 VARIETY DOSA AND JUICE-Malli mane food court	4.1	100.0

91 rows × 3 columns

```
In [46]: fig = px.bar(
    data_frame=df_Expensive_Restaurants,
    x=df_Expensive_Restaurants["Shop_Name"][0:15],
    y=df_Expensive_Restaurants["Cost_for_Two"][0:15],
    color=df_Expensive_Restaurants["Rating"][0:15],
    labels={
        "x": "Restaurant_Name",
        "y": "Approx. Cost_for_Two (₹)",
        "color": "Rating",
    },
)
fig.update_layout(
    template="plotly_dark",
    title="Top 15 Expensive & Highest Rated Restaurants with Approx. Cost for 2 People",
)
fig.show()
```



Cuisine Analysis:

```
In [47]: df_Swiggy["Cuisine"] = df_Swiggy["Cuisine"].str.title()  
df_Swiggy["Cuisine"]
```

```
Out[47]: 0          Sweets
1      North Indian, Home Food, Thalís, Combo
2      South Indian, North Indian, Fast Food, Beverag...
3      American, Arabian, Chinese, Desserts, Fast Foo...
4          Chinese, Thai
...
113      Pan-Asian, Chinese, Asian
114      Arabian, Middle Eastern, North Indian, Grill, ...
115      North Indian, Chinese, Seafood, Biryani
116      North Indian
117      Andhra, Biryani
Name: Cuisine, Length: 118, dtype: object
```

```
In [48]: df_Swiggy["Cuisine"].unique()
```

```

Out[48]: array(['Sweets', 'North Indian, Home Food, Thalís, Combo',
'South Indian, North Indian, Fast Food, Beverages, Jain',
'Amerícan, Arabian, Chinese, Desserts, Fast Food, Mughlai, North Indian',
'Chinese, Thai', 'North Indian',
'Fast Food, North Indian, Chinese', 'Italian',
'South Indian, Kerala, Chinese, North Indian',
'Pizzas, Italian, Mexican',
'Chinese, South Indian, North Indian, Desserts, Fast Food, Kerala, Andhra, Beverages, Mughlai, Seafood',
'Desserts', 'Chinese, Andhra, Biryani, Seafood', 'Chinese',
'South Indian, Chinese, Desserts, North Indian',
'Arabian, Fast Food', 'Desserts, Beverages', 'Indian, Rajasthani',
'Desserts, Bakery', 'Chinese, Healthy Food, North Indian',
'North Indian, Chinese, Hyderabadí', 'Fast Food',
'Chinese, Jain, North Indian, South Indian', 'Indian',
'North Indian, South Indian, Chinese',
'Andhra, Biryani, Chinese, Desserts, Fast Food, Seafood, South Indian',
'Amerícan, Fast Food',
'Biryani, Seafood, North Indian, Chinese, Desserts, Andhra, South Indian',
'Snacks, Amerícan', 'South Indian', 'Kerala, South Indian',
'Mexícan', 'North Indian, Chinese, Biryani',
'Turkish, Portuguese, Amerícan', 'Biryani',
'South Indian, Snacks, North Indian, Chinese',
'Desserts, Fast Food, Sweets, Chaat',
'Chinese, South Indian, Andhra, Hyderabadí', 'Pizzas, Fast Food',
'Biryani, Mughlai, South Indian', 'Chinese, Asian',
'North Indian, Chinese, South Indian', 'Italian, Desserts, Pizzas',
'Biryani, Andhra, South Indian',
'Chinese, Continental, Italian, Mediterranean, Thai, Lebanese, Amerícan, Asian, Beverages, Bakery, Biryani, Caf
e, Desserts, Healthy Food, Mexican, North Indian, Salads, Pizzas',
'Pizzas, Chinese, Pastas, Salads, Amerícan, Continental',
'Andhra, Biryani',
'Chinese, South Indian, North Indian, Fast Food',
'Fast Food, Beverages',
'Biryani, South Indian, North Indian, Fast Food, Andhra, Beverages, Mughlai, Seafood, Punjabi, Hyderabadí, Chine
se',
'Beverages, Chinese',
'South Indian, Biryani, Kerala, North Indian, Chinese',
'Kerala, Chinese', 'North Indian, Chinese',
'Arabian, Beverages, Biryani, Chinese, Desserts, North Indian',
'Biryani, Juices, Kebabs', 'Andhra, South Indian',
'Beverages, Cafe, Snacks', 'North Indian, South Indian',
'Turkish, Portuguese, Amerícan, Grill',
'Home Food, Healthy Food, Indian', 'Ice Cream',
', Hyderabadí, Biryani, Indian, South Indian, Andhra, Tandoor',
Punjabi, North Indian, Chinese, Fast Food, Healthy Food, Mughlai, Desserts',

```

```
'American',
'Biryani, Hyderabad, Andhra, North Indian, South Indian',
'Fast Food, Juices, North Indian',
'North Indian, Chaat, Snacks, Fast Food',
'Desserts, Mughlai, Seafood', 'Ice Cream, Desserts',
'Chinese, North Indian', 'Biryani, Kebabs',
'Andhra, Chettinad, Chinese, Mughlai, North Indian',
'Chettinad, South Indian',
'Continental, Indian, Pan-Asian, Oriental',
'North Indian, Biryani', 'Pan-Asian, Chinese, Asian',
'Arabian, Middle Eastern, North Indian, Grill, Seafood, Kerala, Chinese',
'North Indian, Chinese, Seafood, Biryani'], dtype=object)
```

In [4]: `from collections import Counter`

```
x=["hello","how","are","you","hello","how","are","you","now"]
Counter(x)
```

Out[4]: Counter({'hello': 2, 'how': 2, 'are': 2, 'you': 2, 'now': 1})

In [1]: `from collections import Counter`

```
frequency_list = []
```

```
for i in df_Swiggy["Cuisine"].unique():
    frequency_list.append(Counter(i.split(",")))

print(frequency_list)
```

In [49]:

```
freq_dict = {}
for i in df_Swiggy["Cuisine"].unique():
    Cuisines_Lists = i.split(",") #words list
    for Cuisine in Cuisines_Lists:
        Cuisine = Cuisine.lstrip(" ") #remove left space
        if Cuisine in freq_dict:
            freq_dict[Cuisine] = freq_dict[Cuisine] + 1
        else:
            freq_dict[Cuisine] = 1

print(freq_dict)
print("Total records: \t", len(freq_dict))
```

```
{'Sweets': 2, 'North Indian': 32, 'Home Food': 2, 'Thalis': 1, 'Combo': 1, 'South Indian': 23, 'Fast Food': 16, 'Beverages': 9, 'Jain': 2, 'American': 8, 'Arabian': 4, 'Chinese': 35, 'Desserts': 15, 'Mughlai': 7, 'Thai': 2, 'Italian': 4, 'Kerala': 6, 'Pizzas': 5, 'Mexican': 3, 'Andhra': 12, 'Seafood': 8, 'Biryani': 18, 'Indian': 5, 'Rajasthani': 1, 'Bakery': 2, 'Healthy Food': 4, 'Hyderabadi': 5, 'Snacks': 4, 'Turkish': 2, 'Portuguese': 2, 'Chaat': 2, 'Asian': 3, 'Continental': 3, 'Mediterranean': 1, 'Lebanese': 1, 'Cafe': 2, 'Salads': 2, 'Pastas': 1, 'Punjabi': 2, 'Juices': 2, 'Kebabs': 2, 'Grill': 2, 'Ice Cream': 2, 'Tandoor': 1, 'Chettinad': 2, 'Pan-Asian': 2, 'Oriental': 1, 'Middle Eastern': 1}
```

Total Records: 48

```
In [ ]: d1={  
        {"Aman"}
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

Here you can see the different types of cuisine present and the number of shops sell them

```
In [50]: # Extracting Cuisines name and there frequency  
Cuisine = freq_dict.keys()  
freq = freq_dict.values()  
  
df_Cuisine_Analysis = pd.DataFrame()  
  
# Creating a dataframe having two feature --> Cuisine and Count  
df_Cuisine_Analysis["Cuisine"] = Cuisine  
df_Cuisine_Analysis["Count"] = freq  
  
df_Cuisine_Analysis
```

Out[50]:

	Cuisine	Count
0	Sweets	2
1	North Indian	32
2	Home Food	2
3	Thalis	1
4	Combo	1
5	South Indian	23
6	Fast Food	16
7	Beverages	9
8	Jain	2
9	American	8
10	Arabian	4
11	Chinese	35
12	Desserts	15
13	Mughlai	7
14	Thai	2
15	Italian	4
16	Kerala	6
17	Pizzas	5
18	Mexican	3
19	Andhra	12
20	Seafood	8
21	Biryani	18
22	Indian	5
23	Rajasthani	1

Loading [MathJax]/extensions/Safe.js

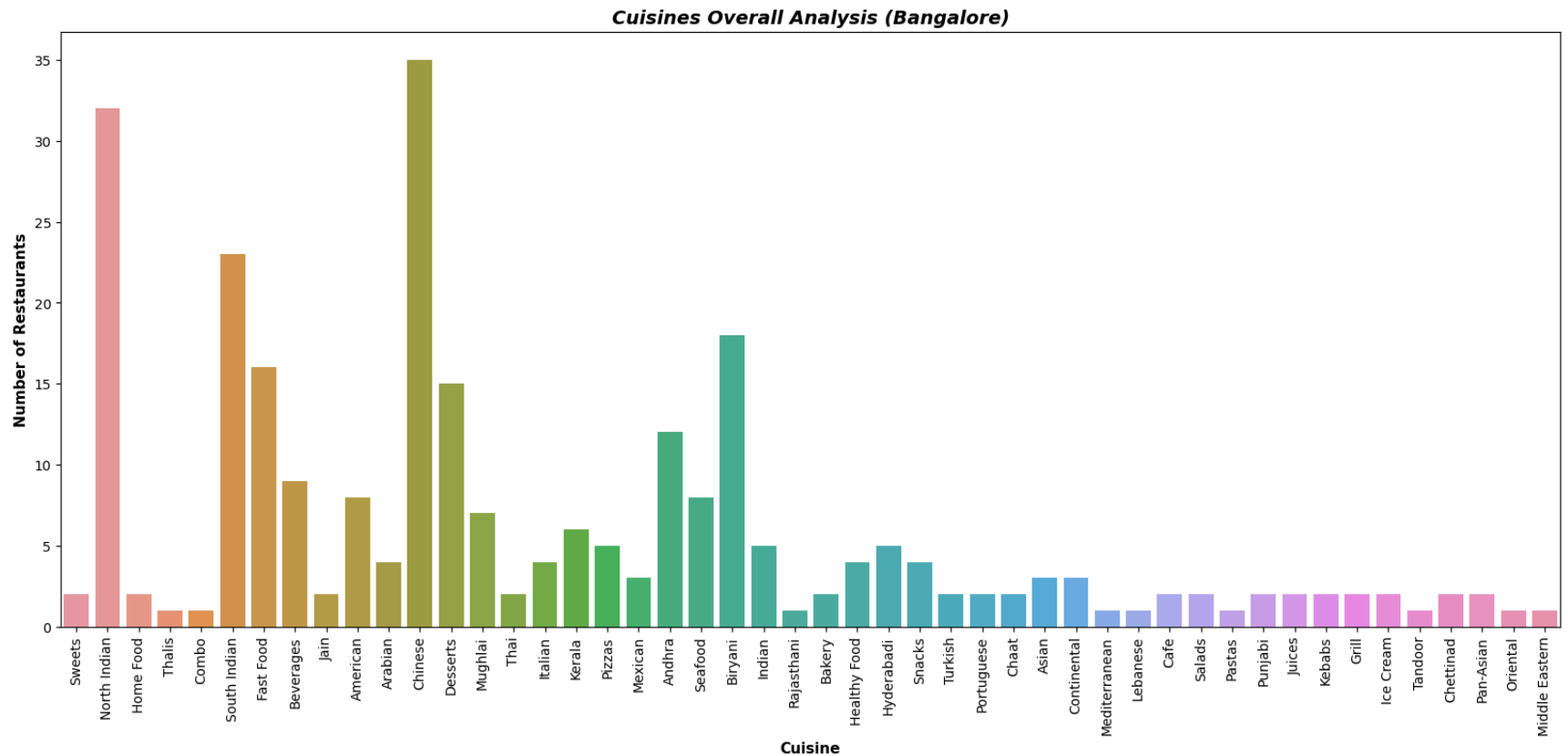
2

	Cuisine	Count
25	Healthy Food	4
26	Hyderabadi	5
27	Snacks	4
28	Turkish	2
29	Portuguese	2
30	Chaat	2
31	Asian	3
32	Continental	3
33	Mediterranean	1
34	Lebanese	1
35	Cafe	2
36	Salads	2
37	Pastas	1
38	Punjabi	2
39	Juices	2
40	Kebabs	2
41	Grill	2
42	Ice Cream	2
43	Tandoor	1
44	Chettinad	2
45	Pan-Asian	2
46	Oriental	1
47	Middle Eastern	1

```

x=df_Cuisine_Analysis["Cuisine"],
y=df_Cuisine_Analysis["Count"],
data=df_Cuisine_Analysis,
)
plt.xticks(rotation=90)
plt.title(
    "Cuisines Overall Analysis (Bangalore)",
    fontsize=14,
    fontweight="bold",
    fontstyle="italic",
)
plt.xlabel("Cuisine", fontsize=11, fontweight="bold")
plt.ylabel("Number of Restaurants", fontsize=11, fontweight="bold")
plt.show()

```



```

In [52]: fig = px.pie(
    data_frame=df_Cuisine_Analysis,
    names=df_Cuisine_Analysis["Cuisine"][:10],

```

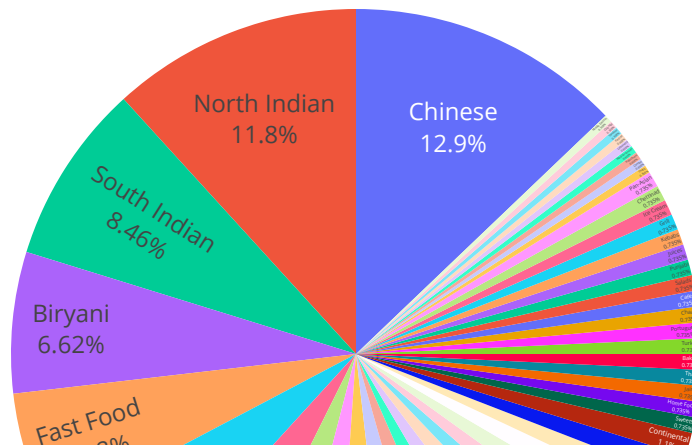
Loading [MathJax]/extensions/Safe.js

```

values = df_Cuisine_Analysis["Count"][:10],
title="Distribution of Cuisines in Bangalore Restaurants",width=700,height=700
)
#fig.update_traces(textposition="inside", textinfo="percent+Label")
fig.show()

```

Distribution of Cuisines in Bangalore Restaurants



Conclusion:

- From the above Visualizations, We can say, **Most of the Restaurants** sell "**Chinese**" which is around '**12.9%**' followed by "**North Indian**" & "**South Indian**" Cuisines which are around '**11.8%**' & '**8.46%**'.

- So, We can also infer that Most of the people are fond of these Cusines.

Area-wise Analysis:

BTM Area

```
In [53]: swiggy_BTMT["Cuisine"].unique()
```

```
Out[53]: array(['Fast Food, North Indian, Chinese', 'Italian',
        'Chinese, South Indian, North Indian, Desserts, Fast Food, Kerala, Andhra, Beverages, Mughlai, Seafood',
        'North Indian', 'Desserts, Bakery',
        'North Indian, Chinese, Hyderabadi', 'Indian',
        'Andhra, Biryani, Chinese, Desserts, Fast Food, Seafood, South Indian',
        'Kerala, South Indian', 'North Indian, Chinese, Biryani',
        'Biryani', 'South Indian, Snacks, North Indian, Chinese',
        'Desserts, Fast Food, Sweets, Chaat',
        'Chinese, South Indian, Andhra, Hyderabadi',
        'North Indian, Chinese, South Indian',
        'Biryani, Andhra, South Indian', 'Fast Food, Beverages',
        'Beverages, Chinese', 'Kerala, Chinese', 'North Indian, Chinese',
        'Arabian, Beverages, Biryani, Chinese, Desserts, North Indian',
        'North Indian, South Indian', 'Chinese, Thai',
        'Chinese, Hyderabadi, Biryani, Indian, South Indian, Andhra, Tandoor',
        'Punjabi, North Indian, Chinese, Fast Food, Healthy Food, Mughlai, Desserts',
        'Biryani, Hyderabadi, Andhra, North Indian, South Indian',
        'Chinese, North Indian',
        'Andhra, Chettinad, Chinese, Mughlai, North Indian',
        'North Indian, Biryani',
        'Arabian, Middle Eastern, North Indian, Grill, Seafood, Kerala, Chinese'],
        dtype=object)
```

```
In [54]: freq_BTMT = {}
for i in swiggy_BTMT["Cuisine"].unique():
    Cuisine_List = i.split(",")
    for Cuisine in Cuisine_List:
        Cuisine = Cuisine.lstrip()
        if Cuisine in freq_BTMT:
            freq_BTMT[Cuisine] = freq_BTMT[Cuisine] + 1
        else:
            freq_BTMT[Cuisine] = 1
```

```
print()
print(len(freq_BT))
```

```
{'Fast Food': 6, 'North Indian': 16, 'Chinese': 18, 'Italian': 1, 'South Indian': 10, 'Desserts': 6, 'Kerala': 4, 'Andhra': 7, 'Beverages': 4, 'Mughlai': 3, 'Seafood': 3, 'Bakery': 1, 'Hyderabadi': 4, 'Indian': 2, 'Biryani': 8, 'Snacks': 1, 'Sweets': 1, 'Chaat': 1, 'Arabian': 2, 'Thai': 1, 'Tandoor': 1, 'Punjabi': 1, 'Healthy Food': 1, 'Chettinad': 1, 'Middle Eastern': 1, 'Grill': 1}
```

26

```
In [55]: Cuisine = freq_BT.keys()
freq = freq_BT.values()

dict_BT = {"Cuisine": Cuisine, "Count": freq}

df_Cuisine_BT = pd.DataFrame(dict_BT)
df_Cuisine_BT.head()
```

```
Out[55]:
```

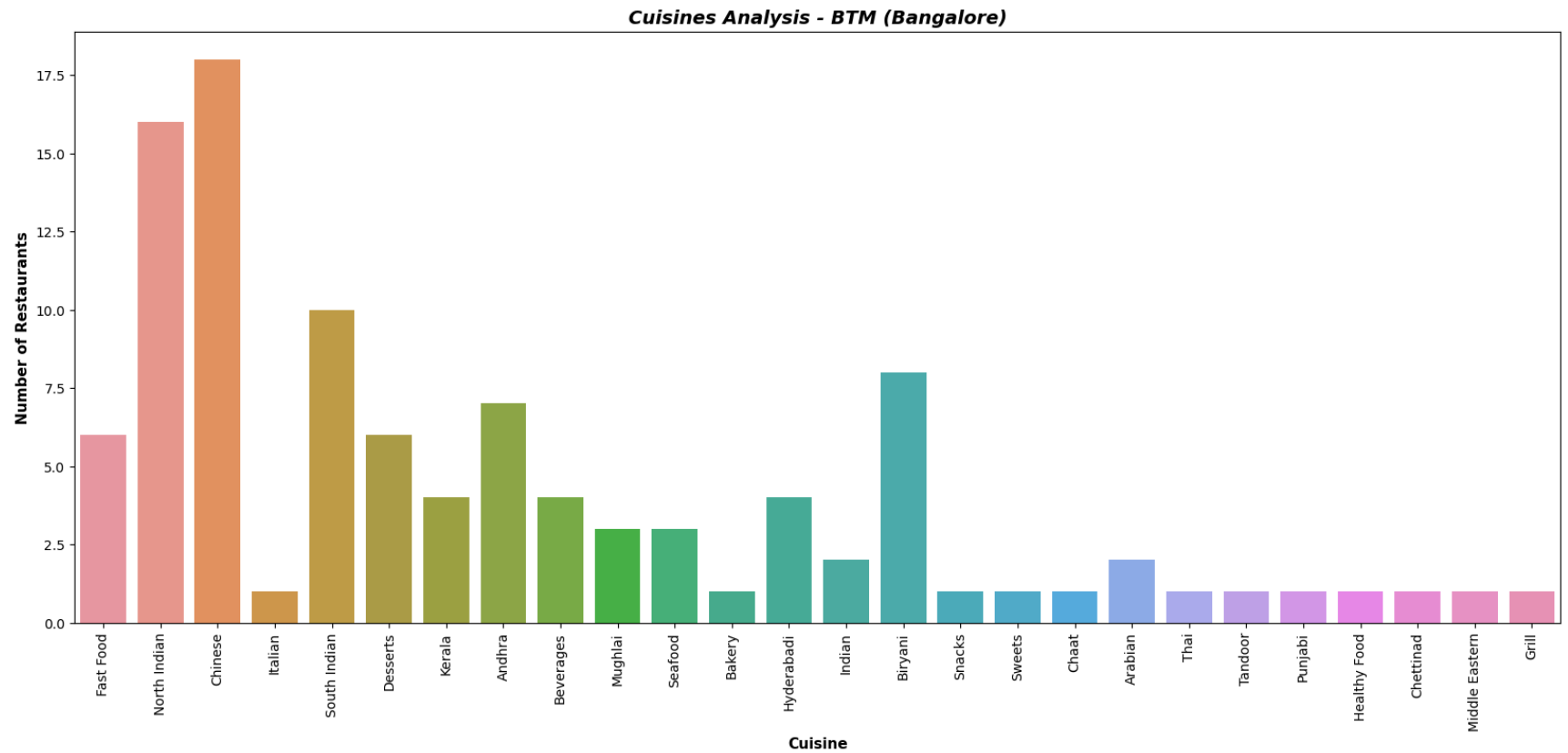
	Cuisine	Count
0	Fast Food	6
1	North Indian	16
2	Chinese	18
3	Italian	1
4	South Indian	10

```
In [56]: plt.figure(figsize=(20, 8))
sns.barplot(
    x=df_Cuisine_BT["Cuisine"], y=df_Cuisine_BT["Count"], data=df_Cuisine_BT
)

plt.xticks(rotation=90)

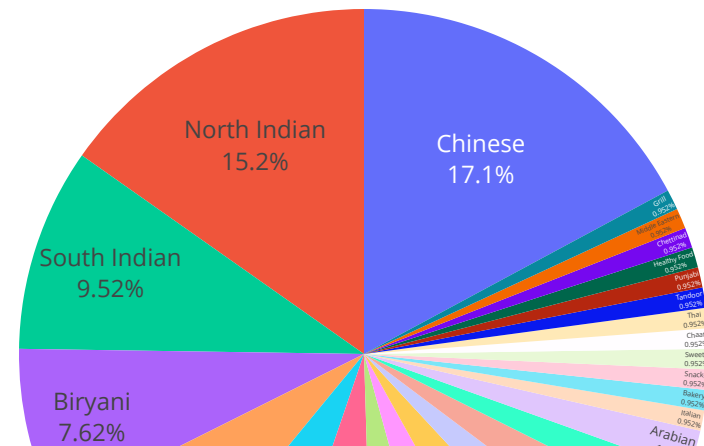
plt.title(
    "Cuisines Analysis - BTM (Bangalore)",
    fontsize=14,
    fontweight="bold",
    fontstyle="italic",
)

plt.xlabel("Cuisine", fontsize=11, fontweight="bold")
plt.ylabel("Number of Restaurants", fontsize=11, fontweight="bold")
plt.show()
```



```
In [57]: fig = px.pie(  
    data_frame=df_Cuisine_BTM,  
    names=df_Cuisine_BTM["Cuisine"],  
    values=df_Cuisine_BTM["Count"],  
    title="Distribution of Cuisines in BTM Bangalore Restaurants",  
)  
  
fig.update_traces(textposition="inside", textinfo="percent+label")  
fig.show()
```

Distribution of Cuisines in BTM Bangalore Restaurants



Conclusion:

- From the above Visualizations, We can say, In **BTM** Area, **Most of the Restaurants** sell "**Chinese**" which is around '**17.1%**' followed by "**North Indian**" & "**South Indian**" Cuisines which are around '**15.2%**' & '**9.52%**'.
- So, We can also infer that Most of the people are fond of these Cusines.

HSR Area

```
Out[58]: array(['North Indian, Home Food, Thalís, Combo',  
        'American, Arabian, Chinese, Desserts, Fast Food, Mughlai, North Indian',  
        'South Indian, Kerala, Chinese, North Indian',  
        'Indian, Rajasthani', 'Chinese, Jain, North Indian, South Indian',  
        'North Indian, Chinese, Biryani', 'Turkish, Portuguese, American',  
        'Desserts', 'South Indian', 'Andhra, Biryani', 'Desserts, Bakery',  
        'Biryani, Juices, Kebabs', 'North Indian', 'Biryani',  
        'Snacks, American', 'Chettinad, South Indian'], dtype=object)
```

```
In [59]: freq_HSR = {}  
for i in swiggy_HSR["Cuisine"].unique():  
    Cuisine_List = i.split(",")  
    for Cuisine in Cuisine_List:  
        Cuisine = Cuisine.lstrip()  
        if Cuisine in freq_HSR:  
            freq_HSR[Cuisine] = freq_HSR[Cuisine] + 1  
        else:  
            freq_HSR[Cuisine] = 1  
  
print(freq_HSR)  
print()  
print(len(freq_HSR))
```

```
{'North Indian': 6, 'Home Food': 1, 'Thalís': 1, 'Combo': 1, 'American': 3, 'Arabian': 1, 'Chinese': 4, 'Desserts': 3,  
'Fast Food': 1, 'Mughlai': 1, 'South Indian': 4, 'Kerala': 1, 'Indian': 1, 'Rajasthani': 1, 'Jain': 1, 'Biryani': 4, 'T  
urkish': 1, 'Portuguese': 1, 'Andhra': 1, 'Bakery': 1, 'Juices': 1, 'Kebabs': 1, 'Snacks': 1, 'Chettinad': 1}
```

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```
In [60]: Cuisine = freq_HSR.keys()  
freq = freq_HSR.values()  
  
dict_HSR = {"Cuisine": Cuisine, "Count": freq}  
  
df_Cuisine_HSR = pd.DataFrame(dict_HSR)  
df_Cuisine_HSR.head()
```

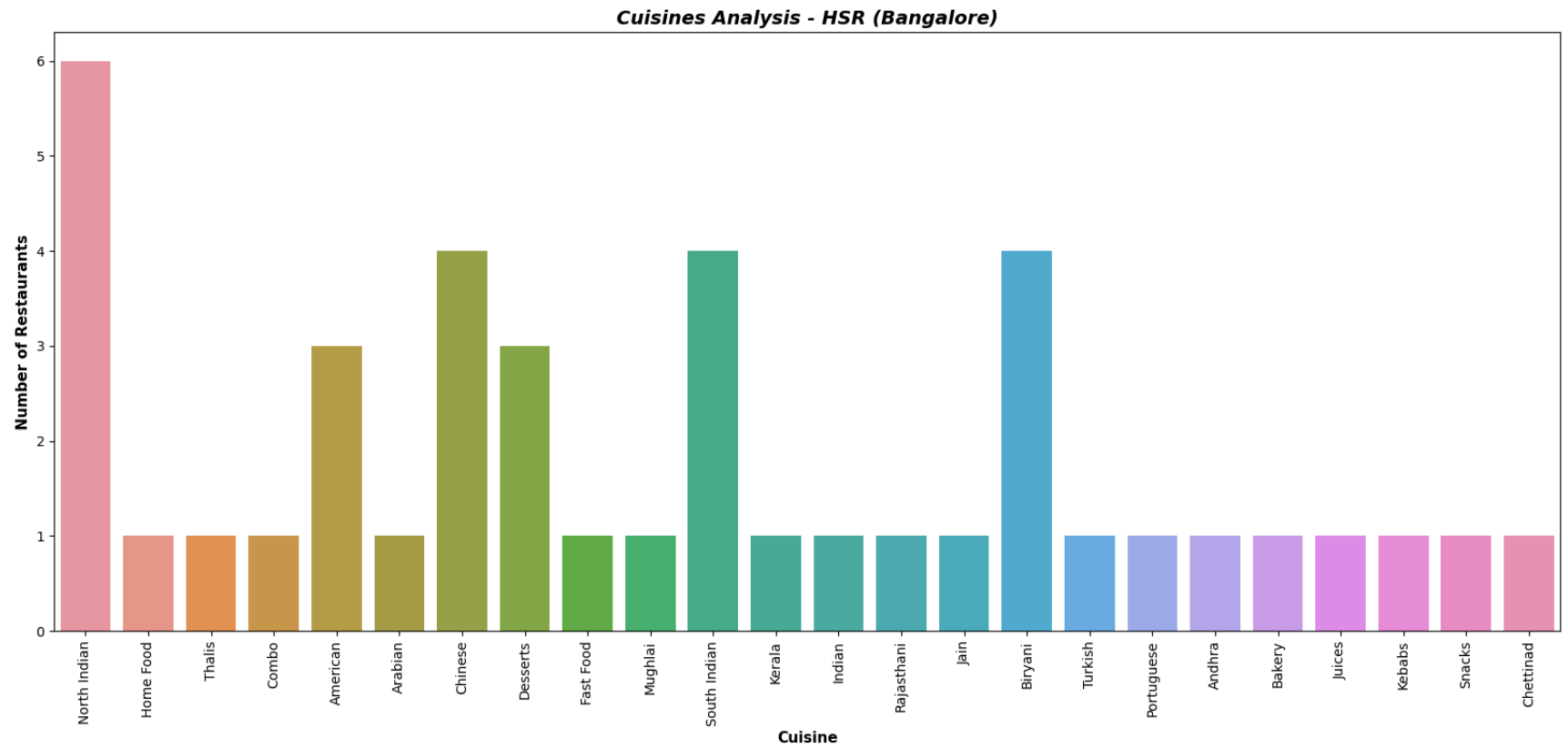
Out[60]:

	Cuisine	Count
0	North Indian	6
1	Home Food	1
2	Thalis	1
3	Combo	1
4	American	3

```
In [61]: plt.figure(figsize=(20, 8))
sns.barplot(
    x=df_Cuisine_HSR["Cuisine"], y=df_Cuisine_HSR["Count"], data=df_Cuisine_HSR
)

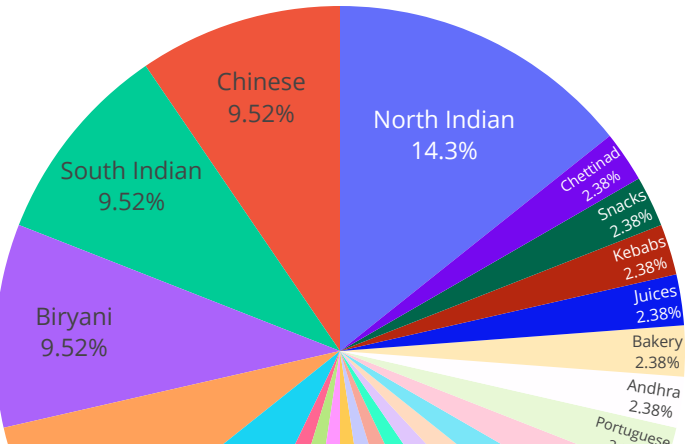
plt.xticks(rotation=90)

plt.title(
    "Cuisines Analysis - HSR (Bangalore)",
    fontsize=14,
    fontweight="bold",
    fontstyle="italic",
)
plt.xlabel("Cuisine", fontsize=11, fontweight="bold")
plt.ylabel("Number of Restaurants", fontsize=11, fontweight="bold")
plt.show()
```



```
In [62]: fig = px.pie(  
    data_frame=df_Cuisine_HSR,  
    names=df_Cuisine_HSR["Cuisine"],  
    values=df_Cuisine_HSR["Count"],  
    title="Distribution of Cuisines in HSR Bangalore Restaurants",  
)  
  
fig.update_traces(textposition="inside", textinfo="percent+label")  
fig.show()
```

Distribution of Cuisines in HSR Bangalore Restaurants



Koramangala Area

```
In [63]: swiggy_Koramangala["Cuisine"].unique()
```

```
Out[63]: array(['Sweets', 'South Indian, North Indian, Fast Food, Beverages, Jain',
               'Chinese, Thai', 'North Indian', 'Pizzas, Italian, Mexican',
               'Desserts', 'Chinese, Andhra, Biryani, Seafood', 'Chinese',
               'South Indian, Chinese, Desserts, North Indian',
               'Arabian, Fast Food', 'Desserts, Beverages',
               'Chinese, Healthy Food, North Indian', 'Fast Food',
               'North Indian, South Indian, Chinese', 'American, Fast Food',
               'Biryani, Seafood, North Indian, Chinese, Desserts, Andhra, South Indian',
               'Snacks, American', 'South Indian', 'Mexican', 'Pizzas, Fast Food',
               'Biryani, Mughlai, South Indian', 'Chinese, Asian',
               'Italian, Desserts, Pizzas',
               'Chinese, Continental, Italian, Mediterranean, Thai, Lebanese, American, Asian, Beverages, Bakery, Biryani, Caf
e, Desserts, Healthy Food, Mexican, North Indian, Salads, Pizzas',
               'Biryani',
               'Pizzas, Chinese, Pastas, Salads, American, Continental',
               'Chinese, South Indian, North Indian, Fast Food',
               'Biryani, South Indian, North Indian, Fast Food, Andhra, Beverages, Mughlai, Seafood, Punjabi, Hyderabad, Chine
se',
               'South Indian, Biryani, Kerala, North Indian, Chinese',
               'Andhra, South Indian', 'Beverages, Cafe, Snacks',
               'Turkish, Portuguese, American, Grill',
               'Home Food, Healthy Food, Indian', 'Ice Cream', 'American',
               'Fast Food, Juices, North Indian',
               'North Indian, Chaat, Snacks, Fast Food',
               'Desserts, Mughlai, Seafood', 'Ice Cream, Desserts', 'Italian',
               'Biryani, Kebabs', 'Continental, Indian, Pan-Asian, Oriental',
               'North Indian, South Indian', 'Andhra, Biryani',
               'Pan-Asian, Chinese, Asian',
               'North Indian, Chinese, Seafood, Biryani'], dtype=object)
```

```
In [64]: freq_Koramangala = {}
for i in swiggy_Koramangala["Cuisine"].unique():
    Cuisine_List = i.split(",")
    for Cuisine in Cuisine_List:
        Cuisine = Cuisine.lstrip()
        if Cuisine in freq_Koramangala:
            freq_Koramangala[Cuisine] = freq_Koramangala[Cuisine] + 1
        else:
            freq_Koramangala[Cuisine] = 1

print(freq_Koramangala)
print()
print(len(freq_Koramangala))
```

```
{'Sweets': 1, 'South Indian': 11, 'North Indian': 14, 'Fast Food': 9, 'Beverages': 5, 'Jain': 1, 'Chinese': 15, 'Thai': 2, 'Pizzas': 5, 'Italian': 4, 'Mexican': 3, 'Desserts': 8, 'Andhra': 5, 'Biryani': 10, 'Seafood': 5, 'Arabian': 1, 'Healthy Food': 3, 'American': 6, 'Snacks': 3, 'Mughlai': 3, 'Asian': 3, 'Continental': 3, 'Mediterranean': 1, 'Lebanese': 1, 'Bakery': 1, 'Cafe': 2, 'Salads': 2, 'Pastas': 1, 'Punjabi': 1, 'Hyderabadi': 1, 'Kerala': 1, 'Turkish': 1, 'Portuguese': 1, 'Grill': 1, 'Home Food': 1, 'Indian': 2, 'Ice Cream': 2, 'Juices': 1, 'Chaat': 1, 'Kebabs': 1, 'Pan-Asian': 2, 'Oriental': 1}
```

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```
In [65]: Cuisine = freq_Koramangala.keys()
freq = freq_Koramangala.values()

dict_Koramangala = {"Cuisine": Cuisine, "Count": freq}

df_Cuisine_Koramangala = pd.DataFrame(dict_Koramangala)
df_Cuisine_Koramangala.head()
```

```
Out[65]:
```

	Cuisine	Count
0	Sweets	1
1	South Indian	11
2	North Indian	14
3	Fast Food	9
4	Beverages	5

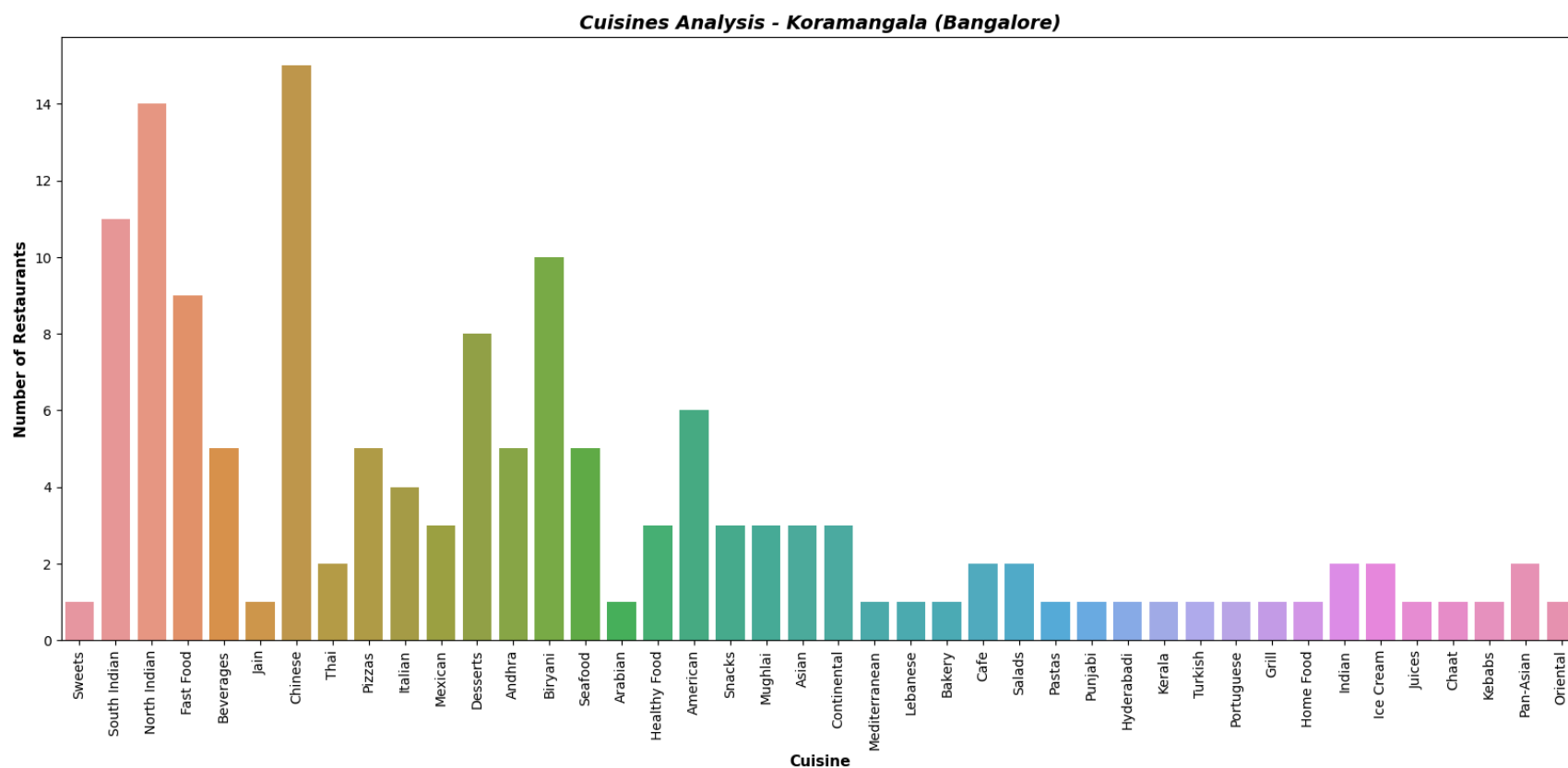
```
In [66]: plt.figure(figsize=(20, 8))
sns.barplot(
    x=df_Cuisine_Koramangala["Cuisine"],
    y=df_Cuisine_Koramangala["Count"],
    data=df_Cuisine_Koramangala,
)

plt.xticks(rotation=90)

plt.title(
    "Cuisines Analysis - Koramangala (Bangalore)",
    fontsize=14,
    fontweight="bold",
    fontstyle="italic",
)

plt.xlabel("Cuisine", fontsize=11, fontweight="bold")
```

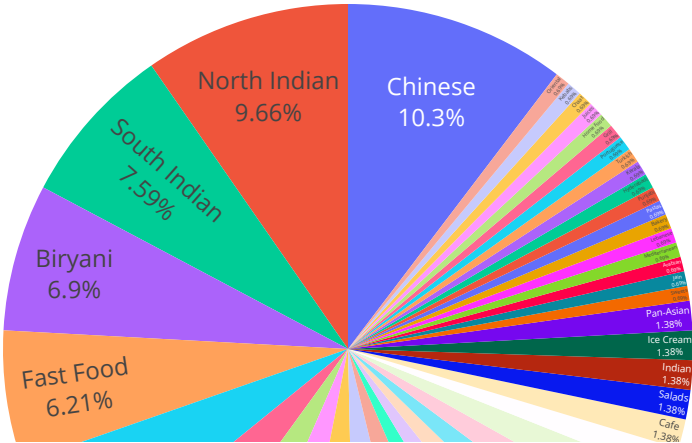
```
plt.ylabel("Number of Restaurants", fontsize=11, fontweight="bold")
plt.savefig('img/Areawise Cuisines Analysis/Koramangala analysis.png')
plt.show()
```



```
In [67]: fig = px.pie(
    data_frame=df_Cuisine_Koramangala,
    names=df_Cuisine_Koramangala["Cuisine"],
    values=df_Cuisine_Koramangala["Count"],
    title="Distribution of Cuisines in Koramangala Bangalore Restaurants",
)

fig.update_traces(textposition="inside", textinfo="percent+label")
fig.show()
```

Distribution of Cuisines in Koramangala Bangalore Restaurants



```
In [ ]: #zomato,prime analysis---assignment
```

```
In [ ]: # pandas,numpy,seaborn
#List,set,tuple,dictionary
#NLP----

#machine wrangling as cleaning
#prediction
```

```
Loading [MathJax]/extensions/Safe.js
#Univariate---
```

```

1.Histplot
2.distributionplot
3.boxplot
#Bivariate
1.Bar Chart
2.Scatter
3.Line Chart

#Categorical
# Univariate---
1.countplot
2.pie chart

#Bivariate
1.Bar
2.Scatter relplot

```

In []:

In []:

In []:

Conclusion:

- From the above Visualizations, We can say, In **Koramangala** Area, "**Chinese**" Cuisines are dominated by around '**10.3%**' followed by "**North Indian**" & "**South Indian**" Cuisines '**9.66%**' & '**7.59%**' Restaurants respectively.
- So, We can also infer that Most of the people are fond of the "**Chinese**" Cuisines.